Fiscal Year:	FY 2023	**Task Last Updated:**	FY 02/04/2023
PI Name:	Baker, John Ph.D.		
Project Title:	Defining the Relationship Between Simulated Weightlessness and Space Radiation on Cardiovascular Disease, and the Effectiveness of a Countermeasure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53226-3548	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	02/04/2019	**End Date:**	02/03/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/03/2024 per NSSC information (Ed., 4/18/23) NOTE: End date changed to 02/03/2023 per NSSC information (Ed., 11/3/22)		
Key Personnel Changes/Previous PI:	December 2022 report: None.		
COI Name (Institution):	Kronenberg, Amy Sc.D. (Lawrence Berkeley National Laboratory) Willey, Jeffrey Ph.D. (Wake Forest University)		
Grant/Contract No.:	80NSSC19K0498		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is one of two proposals selected for funding in response to solicitation NNJ16ZSA001N-SRHHC “Topic 6: Combined Effects of Simulated Weightlessness and Space Radiation on Cardiovascular Structure and Function,” and then descoped and merged at the direction of NASA with a second selected proposal (Principal Investigator-PI Michael Delp, PhD ; Radiation, Simulated Weightlessness and Countermeasures: Effects on Cerebral and Coronary Vascular Function and Structure). This descoped application is designed to address key aspects of the research goals put forward in Topic 6 of NASA Research Announcement (NRA) NNJ16ZSA001N-SRHHC “Space Radiation and Human Health Countermeasures Topics”: 1) to provide detailed physiological and bio-molecular characterization of risk and prediction for cardiac disease from microgravity and space-like radiation doses that are mission relevant for future human spaceflight beyond low Earth orbit and 2) to determine relative biological effectiveness (RBE) values for space radiation in appropriate animals using relevant intermediate as well as late physiological endpoints and effects. These experimentally derived RBE values will provide quantitative inputs into methods and models to calculate degenerative tissue responses in humans following exposures in space. The proposed studies center on assessment of the degenerative tissue effects of space radiation on cardiac disease. The central hypothesis of this project is that simulated weightlessness interacts with whole body exposure to space relevant doses of radiation to potentiate risk for cardiac disease. Three specific aims are proposed to directly address the issue of risk for cardiac disease using experimental approaches in a rat model relevant to the research questions. The proposed investigations directly address important aspects of the NASA Human Research Program (HRP) Degenerative Tissue Gaps, as set forth in the Human Research Roadmap. HRP gaps are identified and addressed by each Specific Aim. The proposed research will provide important information to help close these gaps. Specific Aim A: Simulate a 3-3.5 year deep space human mission in male WAG rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam (SimGCRSim) with total doses of 0.75 Gy and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease. Specific Aim B: Determine the response to photon radiation in male Fischer 344 rats with and without simulated weightlessness to estimate relative biological effectiveness (RBE) for early clinically relevant endpoints of risk for and occurrence of cardiac disease. Rats will be irradiated with 3 doses of X-rays: 3.0 Gy, 8.0 Gy, and 10.0 Gy alongside experimental hindlimb unloaded animals. Specific Aim C: Simulate a 3-3.5 year deep space human mission in male Fischer 344 rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam (SimGCRSim) with total doses of 0.75 Gy and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research benefits life on Earth as it relates to patients who receive therapeutic radiation for the treatment of cancer and to individuals who receive accidental exposure to radiation from a nuclear accident.
Task Progress:	These studies address important aspects of the NASA Human Research Program (HRP) Degenerative Tissue Gaps. To date, combining the spaceflight hazard of reduced weight-bearing via hindlimb unloading (HLU) in rats of any strain with a simplified galactic cosmic ray simulator (SimGCRSim) in conscious animals has not been possible. The aims of our studies are a) to simulate a 3-3.5-year deep space human mission using a rat model and determine the impact of SimGCRSim on the occurrence of cardiac disease and b) determine whether the damage from combined SimGCRSim and HLU is strain dependent. We tested the hypothesis that simulated weightlessness interacts with whole-body exposure to space-relevant doses of radiation to potentiate risk for cardiac disease in two rat strains. Ground-based studies were performed at the NASA Space Radiation Lab (NSRL) at Brookhaven National Lab (BNL) with male WAG/RijCmcr (Wistar) and male Fischer 344 rats ~8.5 months old at the start of the study. SimGCRSim developed by NASA at NSRL comprised of 1000 MeV protons (35%), 600 MeV/n Si ions (1%), 250 MeV/n He ions (18%), 350 MeV/n O ions (6%), 600 MeV/n Fe ions (1%), and 250 MeV protons (39%). Beams were delivered during one irradiation session with rapid beam switching. Dose-rates were maximized based on accelerator output. Rats were HLU for 5 days prior to, during, and 25 days after whole body irradiation with SimGCRSim with a total dose of 1.50 Gy (Wistar) or 0.75 and 1.5 Gy (Fischer 344). Additional Wistar and Fischer 344 rats maintained with normal weight bearing throughout the study (non-HLU rats) received a total dose of 0.75 Gy or 1.5 Gy. Sham-irradiated and sham-HLU rats served as controls. One-month post-irradiation (post-IR), rats were transported to the Medical College of Wisconsin to determine long-term changes in risk for cardiac disease and kidney injury in blood samples withdrawn every 30 days over the 12-month follow-up period. We report the results on the effect of simulated weightlessness combined with space radiation on the occurrence of cardiac disease in male Wistar rats and found that blood urea nitrogen (BUN), an index of kidney injury and blood total cholesterol, a risk factor for cardiac disease in humans, was unchanged for 270-360 days follow-up period in animals irradiated (IR) with 0.75 Gy or 1.5 Gy SimGCRSim without HLU or with HLU + 1.5 Gy SimGCRSim when compared with an appropriate control. Perivascular collagen content in hearts (fibrosis) was determined at the final time-point of this study at 270-360 days post-irradiation. Total body exposure to 0.75 Gy (at 360 days post-IR) or 1.5 Gy (at 270 days post-IR) significantly increased perivascular cardiac collagen content in Wistar rats compared with sham IR age-matched controls with $P = 0.031$ or $P = 0.024$, respectively. Notably, exposure of HLU rats to 1.5 Gy resulted in significant ($P = 0.008$) heart fibrosis compared with sham HLU, age-matched rats evaluated 280 days after irradiation or sham IR. However, the addition of HLU to a 1.5 Gy dose of SimGCRSim that alone results in cardiac fibrosis did not change the degree of cardiac fibrosis beyond the effect of 1.5 Gy alone. The two groups could not be distinguished. BUN and total cholesterol were unchanged in Fischer 344 rats exposed to 0.75 Gy or 1.5 Gy SimGCRSim alone or with HLU when compared with an appropriate control for 270 days follow-up. Heart fibrosis was not detected either after SimGCRSim irradiation alone with a dose of 0.75 Gy or/and 1.5 Gy or with HLU as determined at 270 days follow-up. The base line level of perivascular collagen content in Fischer 344 rats was 1.9x or 1.2x higher when compared with hearts from Wistar rats in Sham IR or Sham HLU group respectively. Our findings provide evidence for the occurrence of cardiac fibrosis resulting from exposure of rats to a 5-ion mixed field beam designed to broadly simulate the composition of the GCR behind shielding, with no additive effect of the simulated spaceflight hazard of reduced weight bearing via HLU on the impact of SimGCRSim using the male Wistar rat model. Male Fischer 344 rats did not show

	any changes in the level of BUN, total cholesterol, or heart fibrosis for 270 days follow-up after SimGCRSim IR alone or in combination with simulated weightlessness.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Abstracts for Journals and Proceedings	Lenarczyk L, Kwok A, Borg A, Willey J, Kronenberg A, Baker JE. "Determining the impact of simulated weightlessness combined with space radiation on the occurrence of cardiac disease in rats." 2022 NASA Human Research Program Investigators' Workshop. 2022 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February , 2022. , Feb-2022
Abstracts for Journals and Proceedings	Lenarczyk M, Alsheikh AJ, Cohen EP, Schaue D, Kronenberg A, Geurts A, Klawikowski S, Mattson D, Baker JE. "T cells contribute to pathological responses in the non-targeted rat heart following irradiation of the kidneys." Toxics. Toxics 2022, 10: 797. , Dec-2022
Abstracts for Journals and Proceedings	Ibrahim EH, Sosa A, Brown SA, An D, Klawikowski S, Baker J, Bergom C. "Myocardial contractility pattern characterization in radiation-induced cardiotoxicity using magnetic resonance imaging: A pilot study with ContractiX." Tomography. 2022; 9:36. , Dec-2022
Articles in Peer-reviewed Journals	Kundu P, Zimmerman B, Perez R, Whitlow CT, Cline JM, Olson JD, Andrews RN, Raber J. "Apolipoprotein E levels in the amygdala and prefrontal cortex predict relative regional brain volumes in irradiated Rhesus macaques." Sci Rep. 2021 Nov 11;11(1):22130. https://doi.org/10.1038/s41598-021-01480-3 ; PubMed PMID: 34764354; PubMed Central PMCID: PMC8585884 , Nov-2021
Articles in Peer-reviewed Journals	Lenarczyk M, Alsheikh AJ, Cohen EP, Schaue D, Kronenberg A, Geurts A, Klawikowski S, Mattson D, Baker JE. "T cells contribute to pathological responses in the non-targeted rat heart following irradiation of the kidneys." Toxics. 2022 Dec 18;10(12):797. https://doi.org/10.3390/toxics10120797 ; PubMed PMID: 36548630; PubMed Central PMCID: PMC9783591 , Dec-2022